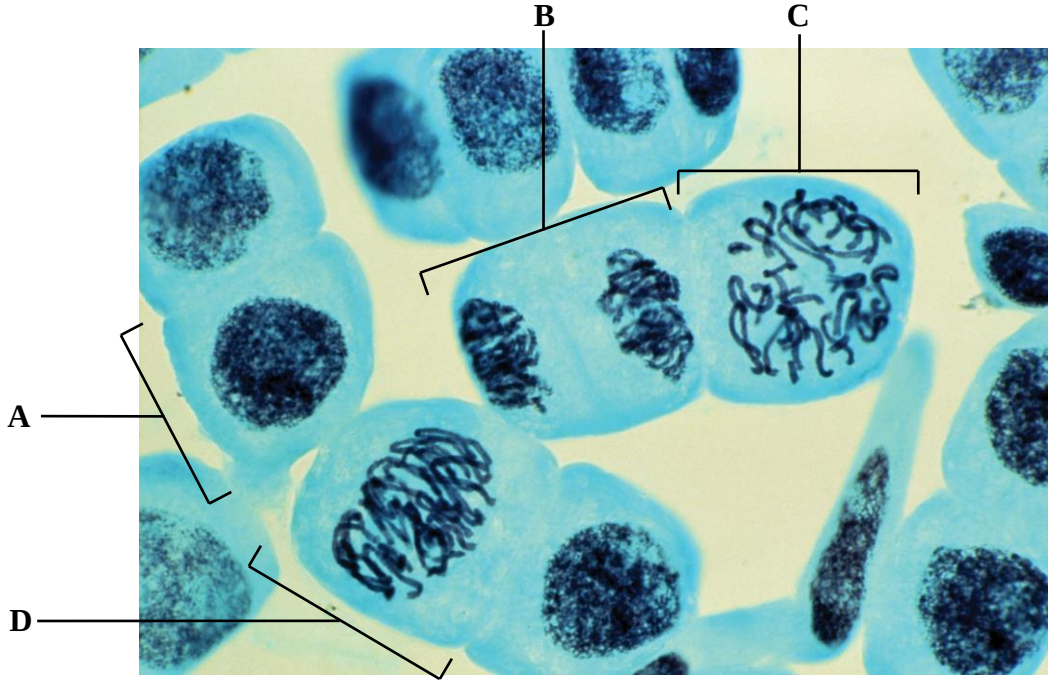
Short questions

|!|BCESB0120000001|!| ★

The photomicrograph below shows some plant cells at different stages of the cell cycle.



Plant cells in various stages of the cell cycle, W.M. (×400)

(a) Using the letters given, arrange the stages of the cell cycle in the correct sequence.
(1 mark)

A → ☐ → ☐ → ☐

(b) State ***three*** events occurring in stage A which prepares the cell for mitosis.(3 marks)

_____

_____

_____

(c) Using the letters given, state the stage with the largest number of chromosomes.
(1 mark)

_____

(d) The number of cells at each stage of the cell cycle was counted and recorded in the table below.

Stage of cell cycle	Number of cells
Interphase	122
Prophase	33
Metaphase	12
Anaphase	7
Telophase	26

One complete cell cycle takes 24 hours. Calculate the length of time spent in prophase. Show your working. (2 marks)

##

(a) A → C → D → B (1)

(b) Replication of DNA (1)

Synthesis of organelles (1)

Synthesis of proteins (1)

(c) Stage B (1)

(d) Total number of cells = 122 + 33 + 12 + 7 + 26 = 200 cells

The length of time spent in prophase = 33 / 200 × 24 hours = 3.96 hours

Step of calculation (1)

Correct answer (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Understanding basic concepts	3
(c)	Applying concepts	1
(d)	Analysing data	1
	Mathematical skills	1

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|!|BCESA0120000002|!| ★★

CCEA GCE AS Biology Assessment Unit AS 1 Jan 2010 Q8(b)

Describe how the behaviour of chromosomes differs during the processes of mitosis and meiosis. Explain the consequences of these differences. (5 marks)

##

Any ***five*** of the following: (5)

- Mitosis involves the separation of the chromatids into new daughter cells. (1)
- The chromosome number of the parent cell is maintained in the daughter cells. (1)
- The daughter cells are genetically identical to the parent cell. (1)
- During prophase I of meiosis, crossing over / chiasmata occur between chromatids of the homologous pair. (1)
- The consequence is that the recombination of genetic make-up on different chromosomes. (1)
- During the first meiotic division, the homologous chromosomes are separated into two daughter cells. (1)
- Since the homologous pairs arrive randomly along the equatorial plane, the chromosomes are independently assorted when subsequently separated. (1)
- In the second division of meiosis the chromatids are separated. (1)
- Meiosis results in the production of haploid cells, which are genetically different. (1)

(5 marks)

##

Skill assessed	Marks
Understanding basic concepts	5

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|!|BCESC0120000003|!|

Complete the table below by stating the number of chromosomes found in each cell.

(4 marks)

	Number of chromosomes in the cell
Red blood cell	(a)
White blood cell	(b)
Sperm	(c)
Nerve cell	(d)

##

(a) 0 (1)

(b) 46 (1)

(c) 23 (1)

(d) 46 (1)

(4 marks)

##

	Skill assessed	**Mark**
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

|!|BCESC0120000004|!|

Complete the following paragraphs with suitable words selected from the list below.

(7 marks)

chromosomes	chromatids	chromatin	spindle fibres
microscope	longer	centromere	DNA
shorter	diploid		

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When a cell is not dividing, the genetic material in the nucleus appears as long and fine threads called (a)_____, which consists of (b)_____ wrapped around special protein molecules.

When the cell is about to divide, the genetic material in the nucleus coils tightly, becoming (c)_____ and thicker. Now they can be seen with a (d)_____ as (e)_____. Each consists of two (f)_____ joined together at a region called (g) _____ .

##

(a) chromatin (1)
(b) DNA (1)
(c) shorter (1)
(d) microscope (1)
(e) chromosomes (1)
(f) chromatids (1)
(g) centromere (1)

(7 marks)

##

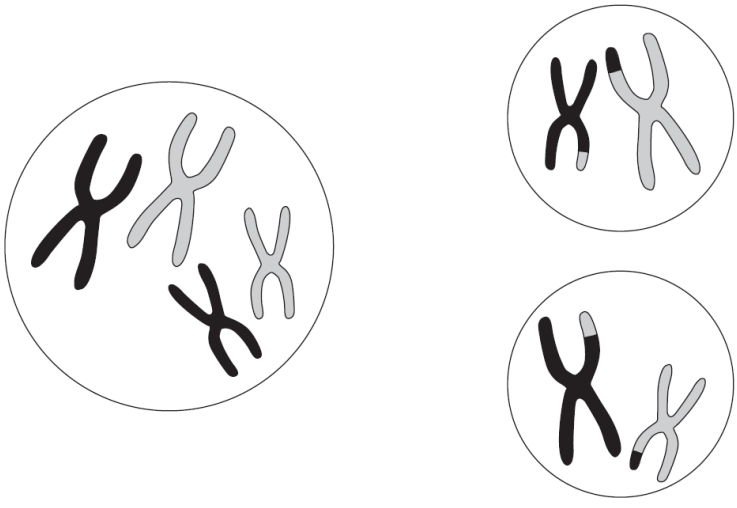
	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1
(g)	Understanding basic concepts	1

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|!|BCESB0120000005|!| ★

CCEA GCE AS Biology Assessment Unit AS 1 Jun 2014 Q2

The diagram below shows two stages in the process of nuclear division by meiosis. The diploid number of the cell shown in stage A is 4.



Stage A **Stage B**

(a) Identify stage B. (1 mark)

(b) Describe the behaviour of the chromosomes between stage A and stage B. (4 marks)

##

(a) Telophase I / Prophase II (1)

(b) Any ***four*** of the following: (4)

- homologous chromosomes pair up (1)
- crossing over occurs / chiasmata form (1)
- homologous pairs line up along the equatorial plane of the cell (1)
- spindle fibres attach to centromeres (1)
- homologous chromosomes are separated / pulled apart (as a result of spindle fibres shortening) (1)

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(5 marks)

##

	Skill assessed	Mark
(a)	Analysing data	1
(b)	Analysing data	4

|!|BCESB0120000006|!| ★

The table below shows some events which may occur during mitosis and/or the first meiotic division. Put a tick (✓) in the appropriate box where the event occurs. Put a cross (×) in the box if the event does not occur. (5 marks)

Event	Mitosis	Meiosis I
(a) Crossing over occurs.		
(b) The number of chromosomes is halved.		
(c) Chromatids separate during anaphase.		
(d) Homologous pairs of chromosomes line up along the equatorial plane of the cell.		
(e) Chromosomes become visible during prophase.		

##

(a) Mitosis: ×; meiosis I: ✓ (1)
(b) Mitosis: ×; meiosis I: ✓ (1)
(c) Mitosis: ✓; meiosis I: × (1)
(d) Mitosis: ×; meiosis I: ✓ (1)
(e) Mitosis: ✓; meiosis I: ✓ (1)

(5 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESB0120000007|!| ★

For each of the stages of meiosis listed in column I, select from column II one sentence that matches it. Put the letter in the space provided. (6 marks)

Column I		***Column II***
Metaphase II	(a)_____	A. Crossing over occurs.
Prophase I	(b)_____	B. Chromosomes line up along the equatorial plane.
Anaphase II	(c)_____	C. Each chromosome becomes visible as two sister chromatids.
Telophase II	(d)_____	D. The sister chromatids move towards opposite poles of the cell.
Anaphase I	(e)_____	E. New nuclear membranes form.
		F. The two members of each homologous pairs move towards opposite poles of the cell.

##

(a) B (1)
(b) A, C (1, 1)
(c) D (1)
(d) E (1)
(e) F (1)

(6 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESB0120000008|!| ★

The table below shows some events which may occur during the first meiotic division and/or the second meiotic division. Put a tick (✓) in the appropriate box where the event occurs. Put a cross (×) in the box if the event does not occur. (5 marks)

Event	Meiosis I	Meiosis II
(a) Crossing over occurs.		
(b) The number of chromosomes is halved.		
(c) Chromatids separate during anaphase.		
(d) Homologous pairs of chromosomes line up along the equatorial plane of the cell.		
(e) Chromosomes become visible during prophase.		

##

(a) Meiosis I: ✓; meiosis II: × (1)

(b) Meiosis I: ✓; meiosis II: × (1)

(c) Meiosis I: ×; meiosis II: ✓ (1)

(d) Meiosis I: ✓; meiosis II: × (1)

(e) Meiosis I: ✓; meiosis II: ✓ (1)

(5 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESC0120000009|!|

Study the following statements. State whether it refers to mitotic cell division or meiotic cell division. (4 marks)

(a) formation of four non-identical haploid daughter cells

_____

(b) formation of new cells in the liver

_____

(c) exchange of genetic material between homologous chromosomes

_____

(d) formation of two identical daughter cells

_____

##

(a) Meiotic cell division (1)

(b) Mitotic cell division (1)

(c) Meiotic cell division (1)

(d) Mitotic cell division (1)

(4 marks)

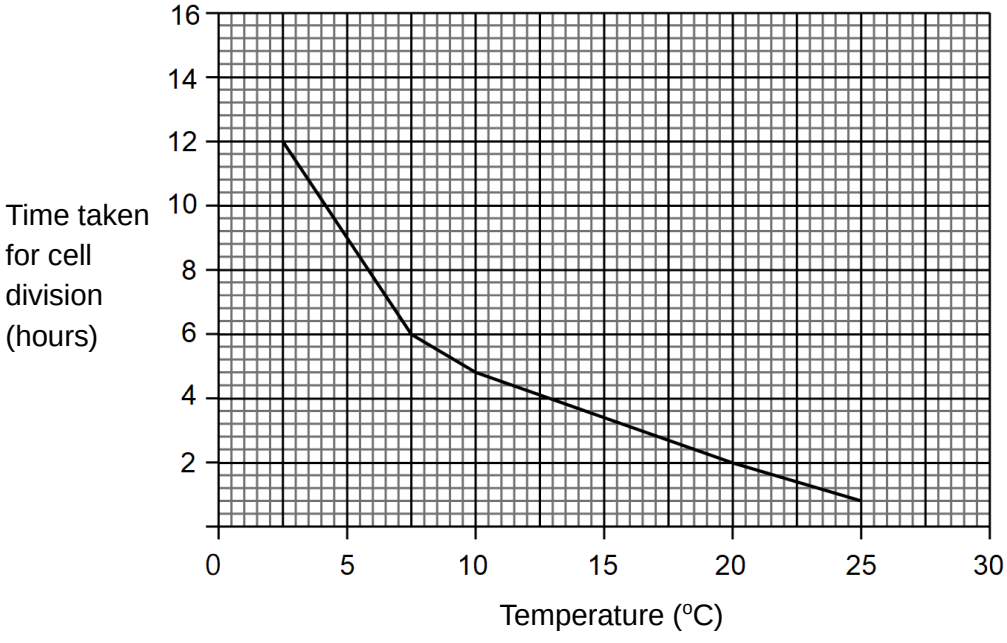
##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCESB0120000010|!| ★

The graph below shows the time taken for cells to divide by mitotic cell division at different temperatures.



(a) What is the effect of temperature on the time taken for cell division? (2 marks)

(b) Starting with one cell during mitosis, at 20 °C, how many cells would there be after:

(i) 2 hours (1 mark)

(ii) 8 hours (1 mark)

##

(a) When the temperature increases, (1)
the time taken for cell division decreases. (1)

(b) (i) 2 (1)

(ii) 16 (1)

(4 marks)

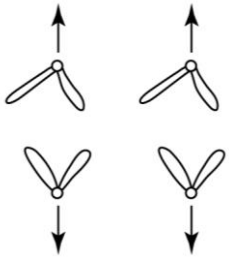
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##

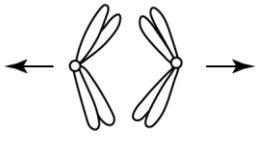
	Skill assessed	Mark
(a)	Analysing data	2
(b)(i)	Mathematical skills	1
(b)(ii)	Mathematical skills	1

|!|BCESB0120000011|!| ★

The diagram below shows a pair of homologous chromosomes at two different stages of meiosis.



Cell X



Cell Y

(a) Name the stages of meiosis shown in the diagram. (2 marks)

_____

_____

(b) State the difference in the behaviours of chromosomes between the two stages. (2 marks)

_____

_____

_____

##

(a) Cell X: Anaphase II (1)

Cell Y: Anaphase I (1)

(b) Cell X: The two sister chromatids of each chromosome are pulled towards opposite poles of the cell. (1)

Cell Y: The two members of a homologous chromosome pair are pulled towards opposite poles of the cell. (1)

(4 marks)

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##

	Skill assessed	Mark
(a)	Analysing data	2
(b)	Analysing data	2

|!|BCESC0120000012|!|
The following paragraph describes the process of mitotic cell division. Complete the paragraph with suitable word(s). (6 marks)

Mitosis is a kind of nuclear division and can be observed using a light microscope. In the first stage, known as (a) _____, the chromosomes become visible. Each chromosome is seen as two sister chromatids attached at the (b)_____. The nuclear (c) _____ breaks down. During the next stage, the chromatids line up along the equatorial plane of the cell and attach to (d) _____. During the next stage known as (e) _____, the sister chromatids separate, one of each pair moving to the opposite pole of the cell. Separate nuclei are formed. The cytoplasm of the cell divides into two equal halves in a process known as (f)_____.

##

(a) prophase (1)
(b) centromere (1)
(c) membrane (1)
(d) spindle fibres (1)
(e) anaphase (1)
(f) cytokinesis (1)

(6 marks)

##

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	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

|!|BCESC0120000013|!|
The following shows the different stages of the cell cycle.

A. Separation of the members of homologous chromosomes

B. Lining up of homologous pairs in the middle of the cell

C. Replication of DNA

D. Separation of sister chromatids

E. Pairing up of homologous chromosomes

(a) Using the letters, arrange the above stages in the correct order. (1 mark)

_____

(b) Give ***three*** differences between mitotic cell division and meiotic cell division. (3 marks)

_____

_____

_____

_____

##

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(a) C, E, B, A, D (1)

(b) Any ***three*** of the following comparisons or other reasonable answers: (1 mark each) (3)

Mitotic cell division	Meiotic cell division
- one nuclear division	- two nuclear divisions
- no pairing of homologous chromosomes	- with pairing of homologous chromosomes.
- occurs in growing areas	- occurs in reproductive organs
- produces body cells	- produces gametes
- genetic materials of daughter cells identical as those of the parent cell	- genetic materials of daughter cells different from those of the parent cells

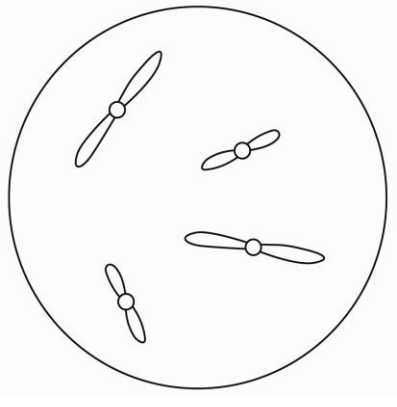
(4 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	3

|!|BCESC0120000014|!|

The following shows a parent cell that contains four chromosomes.



(a) Draw diagrams to show the chromosomes of one of the resulting daughter cells if the parent cell undergoes

(i) mitotic cell division; (1 mark)

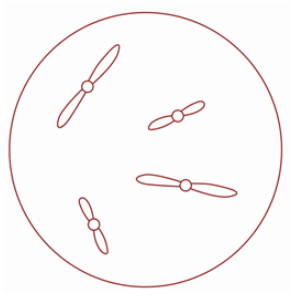
(ii) meiotic cell division. (1 mark)

(b) Describe how meiotic cell division can produce gametes with genetic variation. (3 marks)

##

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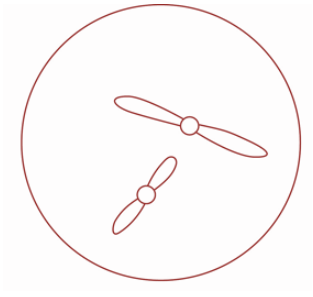
(a) (i)



(1)

Correct drawing

(ii)



(1)

Correct drawing

(b) The independent assortment of the chromosomes in the first meiotic division leads to various genetic combinations in the gametes formed. (1)

Crossing over that may occur during prophase I of meiosis creates further new combinations of genetic make-ups in the gametes. (1)

Random fusion of the gametes increases the genetic variations among the offspring. (1)

(5 marks)

##

	Skill assessed	Mark
(a)(i)	Presenting information	1
(a)(ii)	Presenting information	1
(b)	Understanding basic concepts	3

|!|BCESC0120000015|!|

Complete the following paragraph with suitable words. (5 marks)

Mitosis is the process of nuclear division and it consists of four stages: prophase, (a)_____, (b)_____ and telophase. The daughter cells produced by mitotic cell division are genetically (c)_____ to their parent cell. Mitotic cell division is important in (d)_____ reproduction of some organisms, such as (e)_____.

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##

(a) metaphase (1)
(b) anaphase (1)
(c) identical (1)
(d) asexual (1)
(e) *Amoeba* / *Paramecium* (Or other acceptable answers) (1)

(5 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCESC0120000016|!|
Complete the following paragraph with suitable words. (6 marks)

Meiotic cell division results in the formation of (a)_____ daughter cells, each with (b)_____ the number of chromosomes of the parent cell in all organisms carrying out (c)_____ reproduction. During (d)_____, the nuclei of the male and female gametes fuse and produce a (e)_____. Therefore, the (f)_____ chromosome number can be restored and kept constant generation after generation.

##

(a) four (1)
(b) half (1)
(c) sexual (1)
(d) fertilization (1)
(e) zygote (1)
(f) diploid (1)

(6 marks)

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##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

|!|BCESC0120000017|!|

For each of the stages of mitosis listed in ***Column 1***, select from ***Column 2*** one phrase that matches it. Put the appropriate letter in the space provided. (4 marks)

Column 1			***Column 2***	
Prophase	(a)	_____	A.	Chromosomes line up in the middle of the cell.
Metaphase	(b)	_____	B.	DNA molecules coil into chromosomes.
Anaphase	(c)	_____	C.	DNA replication occurs.
Telophase	(d)	_____	D.	Sister chromatids of each chromosome are pulled apart towards opposite poles of the cell.
			E.	A nuclear membrane forms around each set of daughter chromosomes.

##

(a) B (1)
(b) A (1)
(c) D (1)
(d) E (1)

_____
(4 marks)

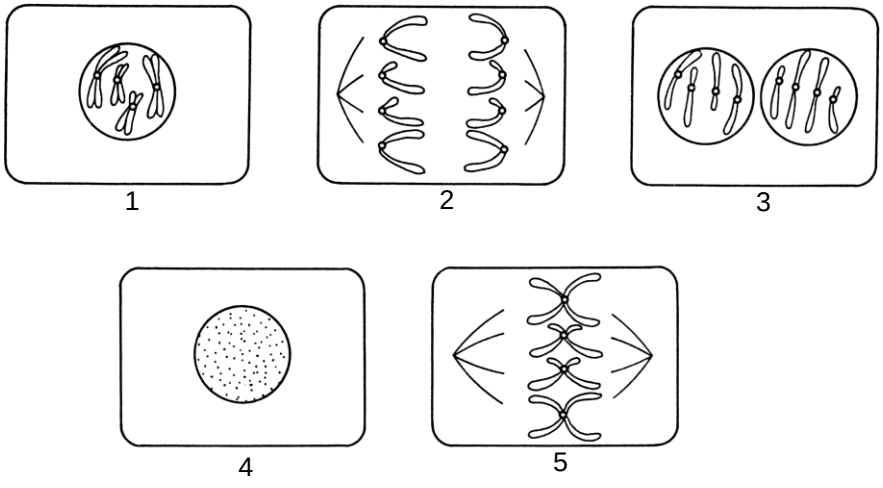
##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCESC0120000018|!|

The following diagrams show five stages in the process of cell division.



(a) Arrange the different stages of cell division in the correct order. (2 marks)

_____

(b) Name this type of cell division. (1 mark)

_____

(c) State where this type of cell division occur in a flowering plant. (1 mark)

_____

##

(a) 4 → 1 → 5 → 2 → 3 (2)

(b) Mitotic cell division (1)

(c) Root tip / shoot tip / apical meristem / lateral meristem (Or other acceptable answers) (1)

(4 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)	Analysing data	1
(c)	Understanding basic concepts	1

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